STEP 1
Order Entry
ERP sales order

STEP 2
Credit & Release
ERP order status

STEP 3
Pick & Pack
WMS

STEP 4
Carrier Tender
TMS

STEP 5
Pickup & Transit
Carrier EDI 214 / visibility

OUTCOME (Y)
Delivered On Time?
Proof of delivery

▲ Queue: awaiting credit/release

▲ Queue: awaiting wave/pick

▲ Queue: awaiting tender

▲ Queue: awaiting pickup

▲ In transit

Decision window: risk score must be available here to expedite, re-sequence, or re-tender

Limited options

Too late to act

DECISION
Can we commit to the requested date?

QUESTIONS
Is the requested date feasible against lead time?

Is the item available to promise?

VARIABLES
- Requested vs. promised date gap S1 T
  - ↳ *ERP order header*
- Quoted lead time S1 T
  - ↳ *ERP item master*
- Available-to-promise qty S1
  - ↳ *ERP inventory and ATP*
- Discount offered S1
  - `Discount_offered`
  - ↳ *ERP pricing*
- Product importance S1
  - `Product_importance`
  - ↳ *Product master*
- Product cost S1
  - `Cost_of_the_Product`
  - ↳ *ERP item cost*

Primary system: ERP order header and ATP

DECISION
Release the order now or hold it?

QUESTIONS
How long has the order been waiting?

How much slack remains to the promise date?

VARIABLES
- Order-to-release hours S2 T
  - ↳ *ERP status log*
- Slack days to promise date S2 T
  - ↳ *ERP order header*
- Credit hold flag and duration S2 T
  - ↳ *ERP credit module*
- Reason code for hold S2
  - ↳ *Not recorded; email or notes*

Primary system: ERP status change log

DECISION
Prioritize this order in the wave?

QUESTIONS
Is the warehouse running behind?

Is the item difficult to handle or ship?

VARIABLES
- Pick backlog at release S3 T
  - ↳ *WMS task queue*
- Labor utilization by shift S3
  - ↳ *Shift labor spreadsheet*
- Pick and pack complete timestamps S3 T
  - ↳ *WMS scan events*
- Shipment weight S3
  - `Weight_in_gms`
  - ↳ *WMS pack station*
- Warehouse zone S1
  - `Warehouse_block`
  - ↳ *WMS location master*

Primary system: WMS task and labor records

DECISION
Expedite, switch carrier, or warn the customer?

QUESTIONS
Will the carrier accept the load on time?

Is this lane running late?

VARIABLES
- Carrier tender acceptance rate S1
  - ↳ *TMS tender history*
- Lane on-time rate, last 30 days S1
  - ↳ *Carrier scorecard spreadsheet*
- Hours from tender to required pickup S4 T
  - ↳ *TMS appointments*
- Shipment mode S1
  - `Mode_of_Shipment`
  - ↳ *Order header*

Primary system: TMS tender and scorecard

DECISION
Intervene while in transit?

QUESTIONS
Did pickup happen on time?

Is the ETA slipping?

VARIABLES
- Pickup delay vs. appointment S5 T
  - ↳ *Dock appointment log*
- Milestone ETA variance S5 T
  - ↳ *Carrier portal or EDI 214*
- Weather and congestion on route S5
  - ↳ *External data feed*

Primary system: carrier milestones, visibility platform

TARGET
Shipment late (1) or on time (0)

VARIABLE
- Target
  - `Reached.on.Time_Y.N`
  - ↳ *Proof of delivery*

Likely recorded during or after delivery. Exclude as predictors.

- Customer rating After
  - `Customer_rating`
  - ↳ *Post-delivery customer survey*
- Care calls After
  - `Customer_care_calls`
  - ↳ *Call center log*

Not mapped to any decision: `Prior_purchases`, `Gender`

In the example dataset · Captured elsewhere in the organization, missing from dataset · Not captured today · Timing violation (post-outcome) · ↳ *Where the data stream lives*

S1 Step where the value is created · T Timing and queue variable

▲ Wait / queue where lateness accumulates

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predictor variables in the dataset

0 of 10
timing and queue variables

2
post-outcome fields to exclude